UNIVERSITÄT PASSAU
Fakultät für Informatik und Mathematik

5820 UE Exercises: Advanced IT-Security



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Repetition

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Identify whether the statements are correct and explain why.

1. If a message is encrypted, its integrity and availability are automatically guaranteed.

False.

2. An insider attacker can also perform an active attack.

True.

3. A system with appropriate safeguards has no remaining security risks.

False.

Repetition

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Identify whether the statements are correct and explain why.

1. If a message is encrypted, its integrity and availability are automatically guaranteed.

False.

Encryption primarily protects confidentiality by preventing unauthorised parties from reading the message.
Integrity: Encryption alone does not necessarily prevent or detect unauthorised modifications.
Availability: An attacker can still block, delete or delay encrypted messages.
Therefore, encryption alone does not guarantee all three CIA properties.

2. An insider attacker can also perform an active attack.

True.

The classifications describe different characteristics of an attacker: Insider: Someone with legitimate access to a system or organisation. Active attacker: Someone who actively interferes with a system, for example by modifying, deleting or injecting data. Example: A university administrator deliberately changes a student's grade without authorisation. This is both an insider attack and an active attack.

3. A system with appropriate safeguards has no remaining security risks.

False.

Safeguards reduce risks, but they generally cannot eliminate all of them.The risk remaining after implementing safeguards is called residual risk.Example: A university regularly backs up its databases. This reduces the potential damage caused by ransomware. However, backups might also fail, become corrupted or be inaccessible during an incident.Therefore, even well-protected systems retain some residual risk.

Task 1 - Cryptography and the Adversary

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Alice and Bob communicate over an insecure channel.

(a) Explain the difference between cryptography, cryptanalysis and cryptology.

(b) Compare the old and new paradigms of cryptography. Why might a trusted third party be necessary?

(c) Explain unconditional and conditional security. What is the fundamental difference?

(d) Alice encrypts her messages. Can an attacker still delete or block them? Which security property would be affected?

Task 1 - Cryptography and the Adversary

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Alice and Bob communicate over an insecure channel.

(a) Explain the difference between cryptography, cryptanalysis and cryptology.

Cryptography: The study of mathematical techniques for protecting information, including confidentiality, integrity and authentication.

Cryptanalysis: The study of techniques for breaking cryptographic systems or defeating their security properties.

Cryptology: The broader field encompassing both cryptography and cryptanalysis.

(b) Compare the old and new paradigms of cryptography. Why might a trusted third party be necessary?

Old paradigm:

- Alice and Bob trust each other.
- They communicate over an insecure channel.
- An external adversary can read, delete or insert messages.
- Cryptography is used to establish a secure logical communication channel.

New paradigm:

- Alice and Bob do not necessarily trust each other.
- Security must also address dishonest behaviour by legitimate participants.
- A Trusted Third Party (TTP) can help settle disputes between participants.

Example: In electronic commerce, a customer might claim that a payment was made, while the merchant claims otherwise. A trusted third party can provide independent evidence or help resolve the dispute.

Task 1 - Cryptography and the Adversary

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Alice and Bob communicate over an insecure channel.

(c) Explain unconditional and conditional security. What is the fundamental difference?

Unconditional security: A cryptographic system remains secure even against an adversary with unlimited computational resources. Its security is analysed using information theory.
Conditional security: A cryptographic system could theoretically be broken, but doing so requires computational resources beyond those available to a realistic adversary.
Its security is analysed using complexity theory.
Fundamental difference: Unconditional security does not rely on computational limitations, whereas conditional security does.

(d) Alice encrypts her messages. Can an attacker still delete or block them? Which security property would be affected?

Yes. Encryption protects message confidentiality but does not necessarily prevent an attacker from deleting, blocking or delaying messages. For example, an attacker controlling a network router could discard encrypted packets without knowing their contents.
The primary security property affected is availability, because Bob cannot access the messages when needed. Additionally, if an attacker modifies encrypted messages, integrity may be threatened.
Encryption alone does not necessarily detect such modifications.
Conclusion: Encryption by itself does not guarantee all three CIA properties.

Task 2 - Cryptographic Hash Functions

(a) Explain the following two fundamental properties of hash functions in your own words:

- Ease of computation
- Compression

(b) Define the following security properties of cryptographic hash functions using mathematical notation. Explain the differences between them.

- Pre-image resistance
- Second pre-image resistance
- Collision resistance

Task 2 - Cryptographic Hash Functions

(a) Explain the following two fundamental properties of hash functions in your own words:

Ease of computation: Given a hash function h and an input x, it is computationally easy to calculate h(x).

Compression: A hash function maps inputs of arbitrary length to outputs of fixed length.

(b) Define the following security properties of cryptographic hash functions using mathematical notation. Explain the differences between them.

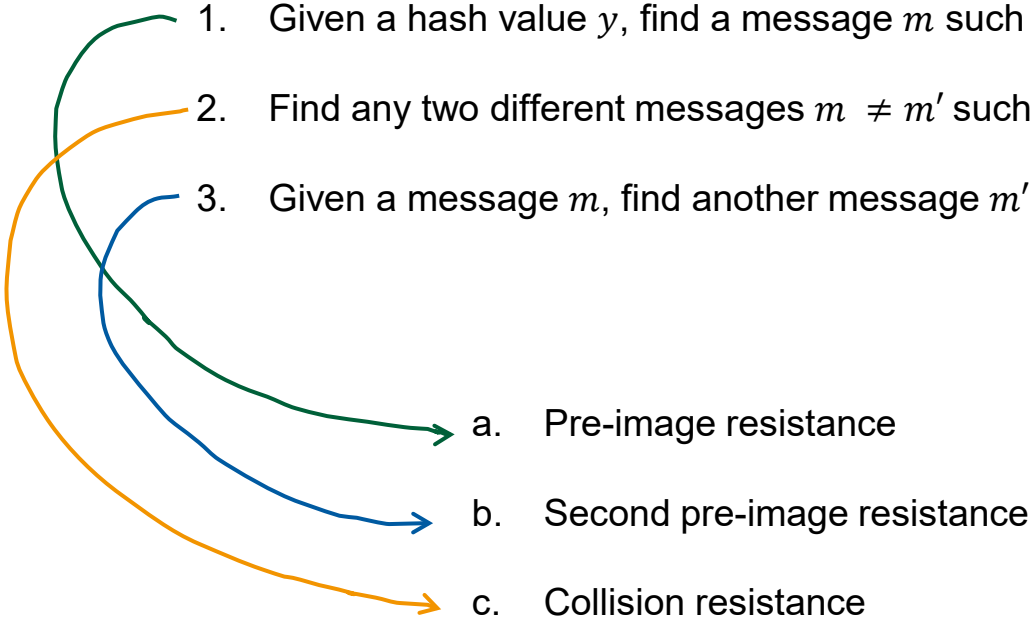
Pre-image resistance: Given a hash value y, it is computationally infeasible to find an input x such that h(x) = y.

Second pre-image resistance: Given an input x, it is computationally infeasible to find another input x' != x such that h(x) = h(x').

Collision resistance: It is computationally infeasible to find any two distinct inputs x != x' such that h(x) = h(x').

Difference: Pre-image resistance starts with a given hash value. Second pre-image resistance starts with a given input. Collision resistance allows the attacker to choose both inputs.

(c) Consider a cryptographic hash function h. Match each of the following situations to the corresponding security property from part (b).



1. Given a hash value y, find a message m such that $h(m) = y$.
2. Find any two different messages $m \neq m'$ such that $h(m) = h(m')$.
3. Given a message m, find another message $m' \neq m$ such that $h(m) = h(m')$.

a. Pre-image resistance

b. Second pre-image resistance

c. Collision resistance

(d) Consider an ideal cryptographic hash function h that produces a hash value of $n = 32$ bits. Assume that hash values are uniformly distributed and independent for randomly selected distinct inputs. How many different hash values can the function produce?

A hash function with n-bit outputs can produce $2^n n$ different hash values.

For n = 32: $2^{32} = 4{,}294{,}967{,}296$ possible hash values.

(d) Explain the difference between a Manipulation Detection Code (MDC) and a Message Authentication Code (MAC). Why is an ordinary cryptographic hash value alone insufficient to authenticate the sender of a message?

MDC: An unkeyed hash function used to detect modifications to data. It provides integrity checking when the reference hash value is trusted.
MAC: A keyed function used to verify message integrity and authenticate the message origin.

An ordinary hash function is publicly computable. An attacker who modifies a message can calculate a new hash value for the modified message. Therefore, an ordinary hash alone cannot authenticate the sender or guarantee integrity against an attacker who can also replace the reference hash.

Task - Find a Collision

$$h(x) = x \bmod 13$$

a) Calculate the hashes of 17, 25, 39, 42, 52.

b) Find two different inputs producing the same hash.

c) Given $h(x) = 5$, find one possible input.

d) Given $x = 17$, find another input with the same hash.

e) Identify which tasks correspond to collision finding, pre-image finding and second pre-image finding.

Why is this function not cryptographically secure, even though it is easy to compute and produces fixed-size outputs?

Task - Hash or MAC?

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What mechanism is sufficient? Choose between an ordinary hash, a MAC, or neither.

Scenario	**Decision**
A downloaded file is compared against a SHA-256 hash obtained from a trusted source.	HASH
Alice sends Bob a message over an insecure channel. Bob wants to verify that it came from someone possessing their shared secret key and was not modified.	MAC
A company publishes a document and its hash value on the same untrusted website.	Neither alone provides trustworthy integrity verification against an attacker controlling the website.
Two users want to communicate confidentially over a public network.	Neither alone, confidentiality requires encryption.

Task - True or False?

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Scenario	
Two different messages can produce the same hash value.	**True.**
A collision-resistant hash function guarantees that collisions do not exist.	**False.**
A MAC can help authenticate a message's origin.	**True.**
Encryption automatically protects against denial-of-service attacks.	**False.**
Pre-image resistance and second pre-image resistance describe the same problem.	**False.**

Task - True or False?

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Scenario	
Two different messages can produce the same hash value.	True. Compression into a fixed-size output makes collisions unavoidable when the input domain is larger.
A collision-resistant hash function guarantees that collisions do not exist.	False. Collision resistance means finding them is computationally infeasible, not that they are impossible.
A MAC can help authenticate a message's origin.	True. A valid MAC provides evidence that someone possessing the secret key generated the tag.
Encryption automatically protects against denial-of-service attacks.	False. An attacker may still block or delete encrypted traffic.
Pre-image resistance and second pre-image resistance describe the same problem.	False. Second pre-image resistance starts with a particular input; pre-image resistance starts with a target hash value.